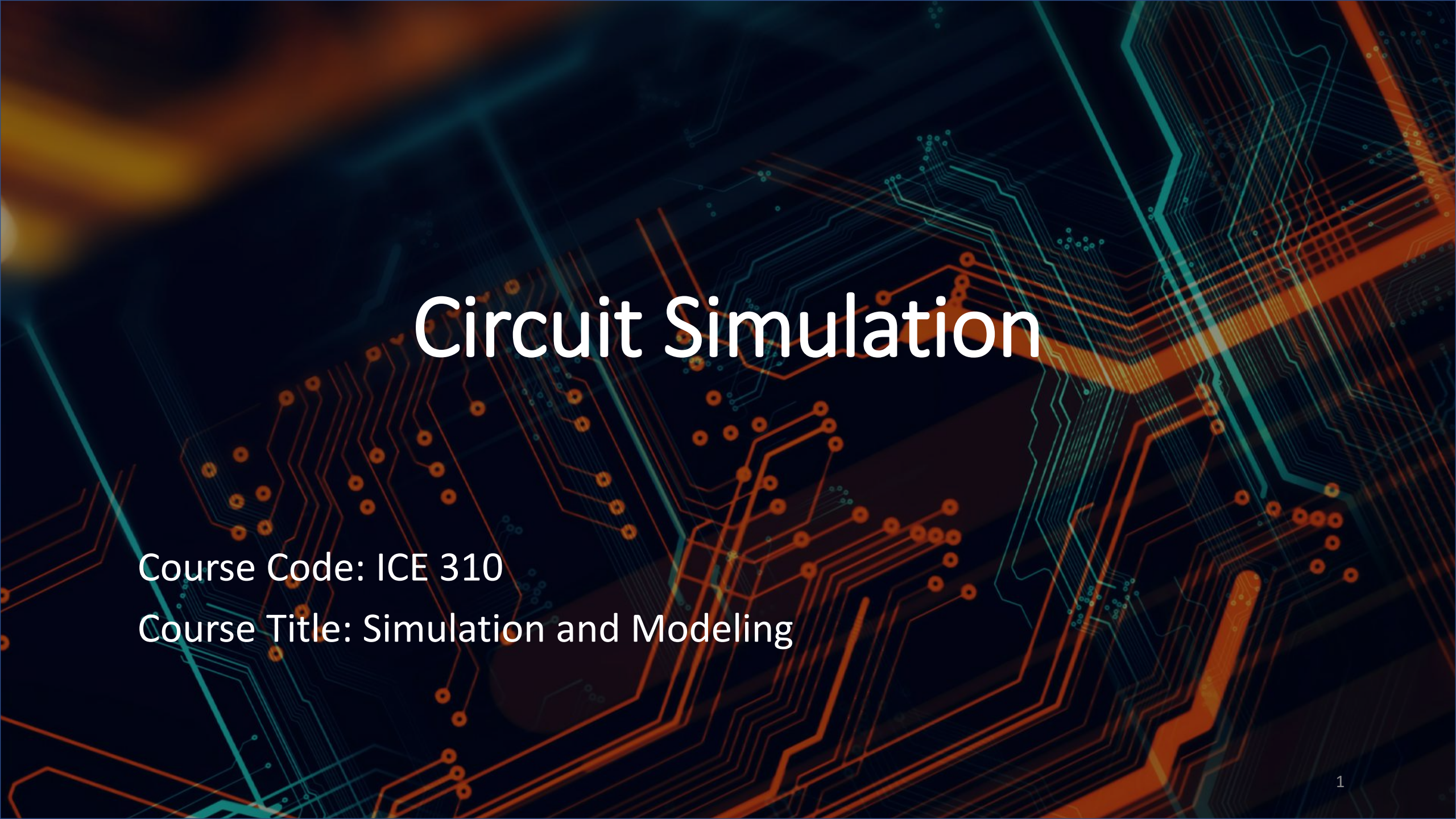




Circuit Simulation

Course Code: ICE 310

Course Title: Simulation and Modeling

What is Circuit Simulation

Circuit simulation is a process in which a model of an electronic circuit is created and analyzed using various software algorithms, which predict and verify the behavior and performance of the circuit.

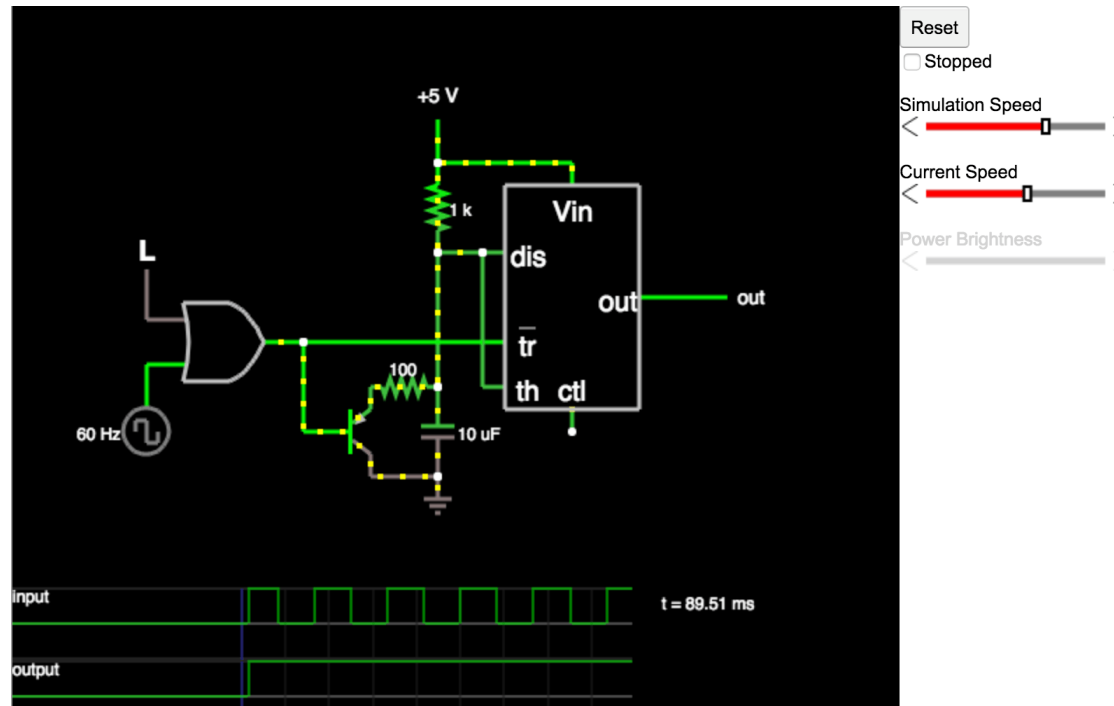


Figure 1: Circuit Simulation software.

Need circuit simulation?

1. Since fabrication of electronic circuits, especially integrated circuits (ICs), is expensive and **time-consuming**, it is faster and more cost-effective to verify the behavior and performance of the circuit using a circuit simulator before fabrication.
2. It is clear that using circuit simulation, we can **test the functionality** of a circuit without actually building the circuit.
3. Circuit simulation is **cost efficient**, time saving and we can easily design complex circuits using the software. This saves design engineers and companies a lot of time and money.
4. Engineers can **reduce the wastage of valuable hardware resources** and proceed to build hardware only after they get satisfactory and accurate results from simulation.

5. We can easily **re-design** the circuit based on the simulation results. At each stage of design and simulation, engineers can verify the performance and compare it to the theoretical responses.
6. Circuit Simulation Software is the **starting point of a new concept or idea**. You could also modify existing projects without designing PCBs.
7. Just like **physical probing and observing waveforms** on an oscilloscope, we can do the same in circuit simulation. You can easily probe different points in a circuit and view the virtual waveforms.

Advantages of Circuit Simulation

1. Most of the popular circuit simulation software such as SPICE, LTspice etc. are **free**. There are some proprietary tools but even those are developed using SPICE algorithm.
2. You can easily **change the models and overall behavior** of the simulator on the go. This helps you in **iterating multiple levels** of simulation to explore the limitations of the circuit.
3. **Changing the values of components** is also very easy. Apart from component values, you could also test the circuit for different input parameters (voltages and currents).
4. **Testing memory** related timings such as **read and write times** and **latency of memory** modules becomes very easy with dedicated analog simulation.

Advantages of Circuit Simulation

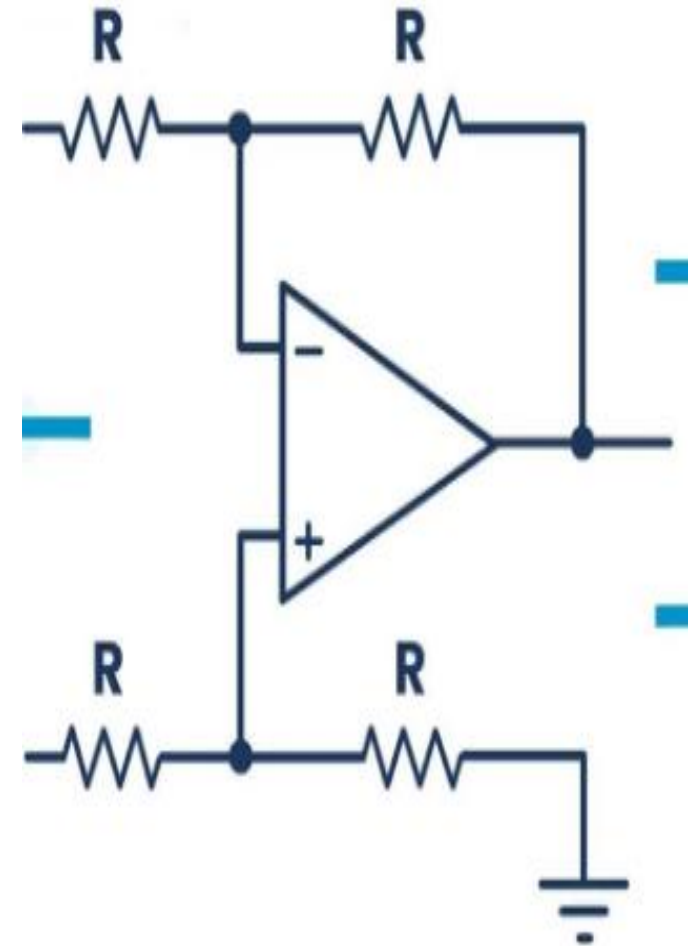
5. You can also use simulators to **analyze noise and cross talk** in complex RF and high-speed designs.
6. Apart from **regular circuit**, **simulators also have tools for testing power supply designs (SMPS)** and power distribution systems.
7. **Digital Simulators check the voltage levels for logic 0 and logic 1** while using Analog Simulators, you can check the duration of the transition between these logic levels.

Types of Circuit Simulation

1. Analog Circuit Simulation
2. Digital Circuit Simulation
3. Mixed-Mode Circuit Simulation

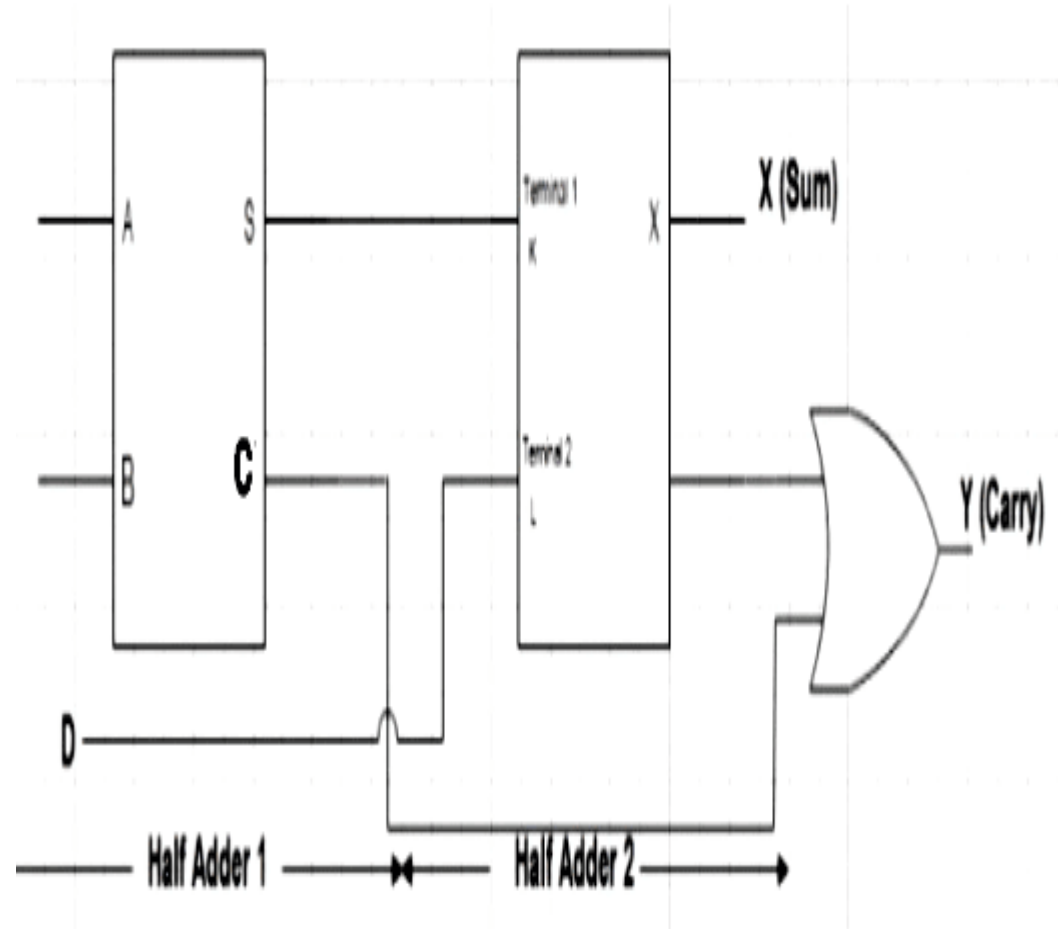
1. Analog Simulation

- In analog simulation, we usually work with analog components and signals.
- In this simulation, the **mathematical models must be highly accurate to run frequency domain (AC), time domain (transient) and non-linear quiescent (DC) modes.**
- **SPICE and FastSPICE** are two popular analog simulators that use very accurate linear and non-linear component models to analyze a circuit's behavior.



2. Digital Simulation

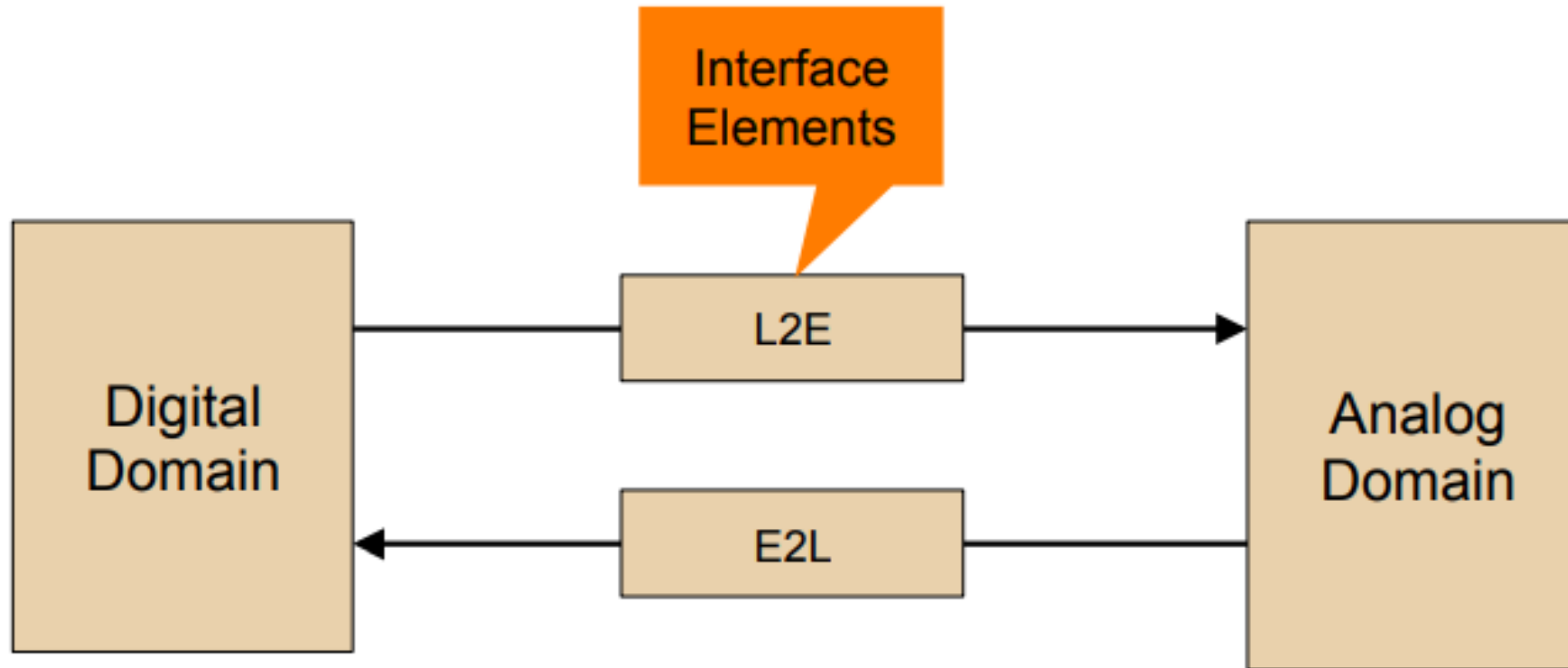
- In digital simulation, work with **two discrete voltage levels**.
- In digital electronics, we call them as Logic 0 and Logic 1.
- HDL such as Verilog **and VHDL** are the backbone of digital simulation.



3. Mixed-Mode Simulation

- It is a combination of both analog and digital simulations.
- This type of simulation is very useful in mixed circuit system design and analysis.
- With the mixed-mode simulation, the analog and digital parts are separate entities with respective tools and resources.

3. Mixed-Mode Simulation



Levels of Simulation

- **Block-Level Simulation:** In this simulation, we divided different parts of several functional blocks and perform simulation on these blocks as a whole. This is very important is a modular design and also helps in reusing the existing functional blocks in other circuits as well.
- **Chip-Level Simulation:** As the name suggests, in a chip-level simulation, we take an integrated circuit (a chip) and simulate its functionality as a single entity. This ensures that the chip is working as intended.

- **Memory Simulation:** Memory is an important part of modern computer systems (large or small). Hence, there are dedicated tools and simulators to test **the memory timings very accurately**. You can simulate SRAM, DRAM, NAND Flash etc.
- **Mixed-Signal Simulation:** Modern **system-on-chips or SOC**s are actually a combination of several analog and digital functional elements. Hence, we can use mixed-signal simulation techniques to see how analog and digital signals behave and interact in a tight space.
- **RF Simulation:** While RF design is still magic to some people, engineers use complex RF simulation tools to test and verify various RF related IC such as **mixers, amplifiers, PLLs, and many more**.

Popular Simulation Software

- SPICE
- ngspice
- LTspice (SPICE)
- PSIM
- Multisim (SPICE)
- Cadence Spectre
- Synopsys PrimeSim (SPICE and FastSPICE)

Sample Question

You are designing a **solar inverter** circuit that converts DC from solar panels into AC for home use. The inverter must produce a stable 230 V, 50 Hz AC output. The solar panel voltage varies with sunlight intensity. The inverter has protection circuits to prevent damage from overloads and short circuits. You want to check if the inverter can maintain a stable AC output when solar input voltage changes.

You want to verify that the protection circuits activate correctly under fault conditions before building hardware. You want to make sure that switching components (like MOSFETs) don't generate harmful voltage spikes that can damage other parts.

- **With respect to it, Explain do you need circuit simulation or not? Why and why not? /What benefits does simulating this inverter circuit bring compared to directly building it and testing?**

Is there any disadvantages?

Reference

1. <https://www.electronicshub.org/circuit-simulation/>
2. <https://www.synopsys.com/glossary/what-is-circuit-simulation.html>